Urban Bike Sharing Performance Dashboard (2022-2025)

1. Project Overview and Objective

This project focuses on analysing bike-sharing system data using **Power BI** to derive meaningful insights about station performance, bike availability, and system utilization.

The main objective is to clean, transform, and visualize the dataset to understand demand patterns, identify underperforming areas, and improve operational efficiency.

2. Data Source

- **Dataset Type:** Bike Sharing System Data
- **Domain:** Transportation / Urban Mobility
- **Data Includes:** Station-level operational data such as bike availability, capacity, and status

3. Problem Statement

- To analyse bike availability trends across cities
- To identify high and low performing cities
- To examine station utilization efficiency
- To detect stations with zero bike availability
- To analyse unused capacity across stations
- To evaluate the impact of banking and bonus features

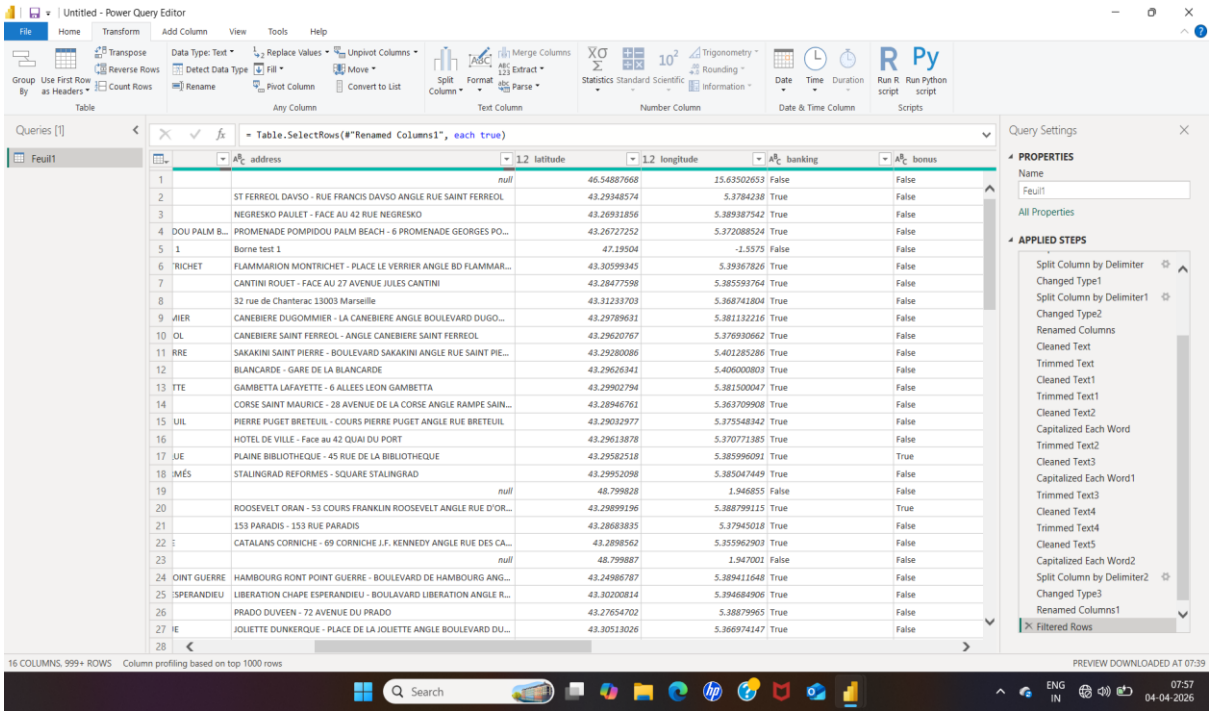
4. Attribute (Column) Details

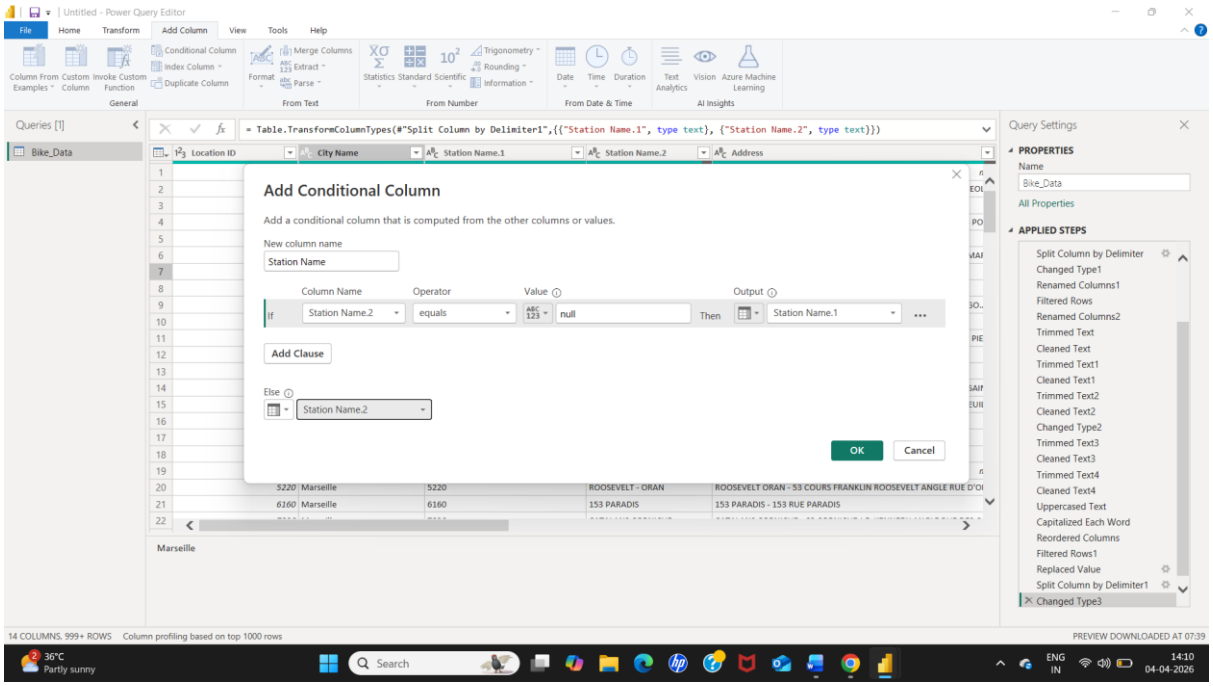
Attribute	Description
Station Name	Name of the bike station
City Name	Location of the station
Available Bikes	Number of bikes available
Available Bike Stands	Number of empty docking points
Bike Stands	Total station capacity
Status	Station status (OPEN / CLOSED)
Banking	Availability of payment facility
Bonus	Availability of bonus/incentives
Latitude	Geographic coordinate
Longitude	Geographic coordinate
Last Update	Timestamp of record

5. Data Cleaning and Transformation

- Removed null and inconsistent values
- Standardized column names
- Ensured correct data types (numeric, text, datetime)

- Removed space by Trim and Clean
- Used Conditional column to rearrange two separated column by delimiter.
- Ensured all data types matches with the columns.

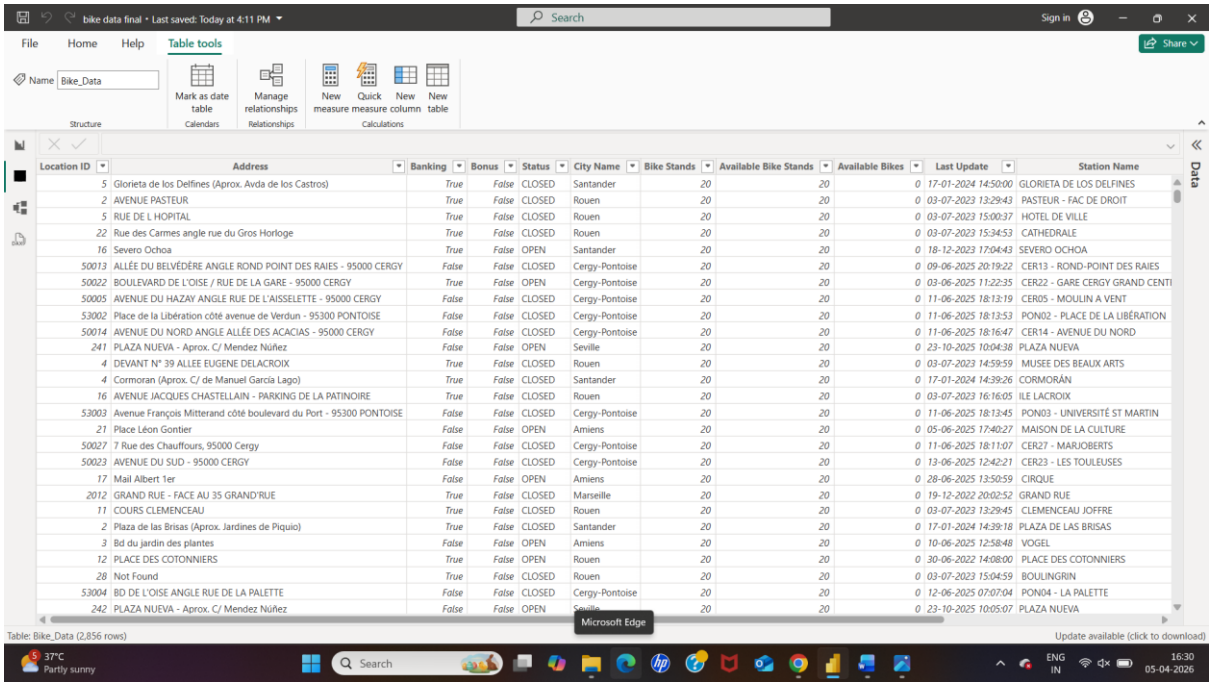




6. Data Modelling and DAX Implementation

Fact Table

Bike_Data (main dataset containing all records)

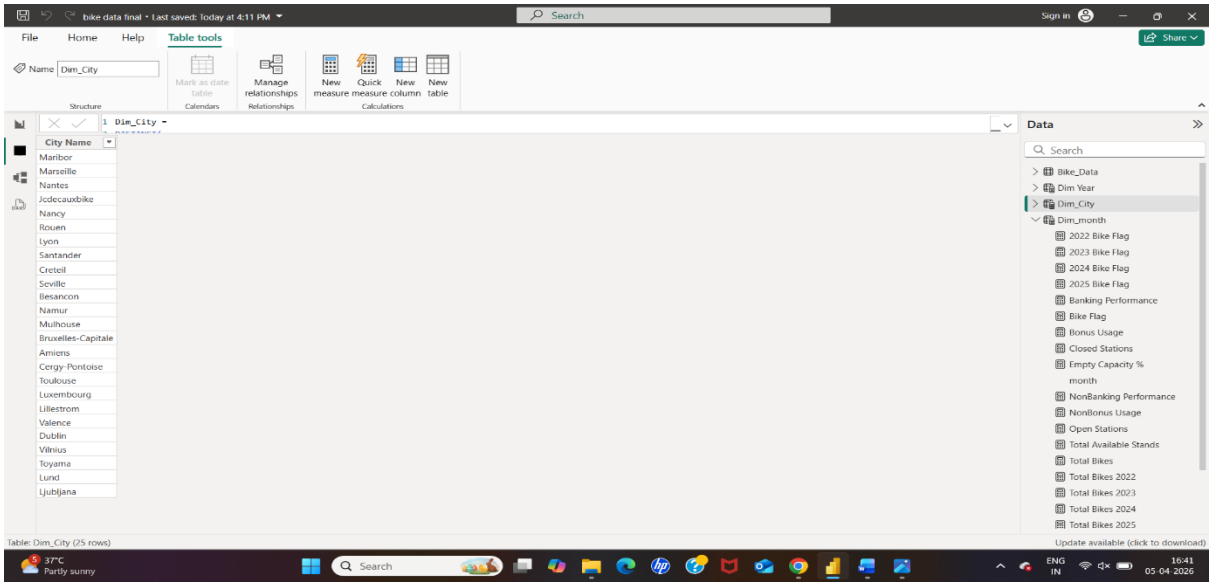


Location ID	Address	Banking	Bonus	Status	City Name	Bike Stands	Available Bike Stands	Available Bikes	Last Update	Station Name
5	Glorieta de los Delfines (Aprox. Avda de los Castros)	True	False	CLOSED	Santander	20	20	0	17-01-2024 14:50:00	GLORIETA DE LOS DELFINES
2	AVENUE PASTEUR	True	False	CLOSED	Rouen	20	20	0	03-07-2023 13:29:43	PASTEUR - FAC DE DROIT
5	RUE DE L HOPITAL	True	False	CLOSED	Rouen	20	20	0	03-07-2023 15:00:37	HOTEL DE VILLE
22	Rue des Carmes angle rue du Gros Horloge	True	False	CLOSED	Rouen	20	20	0	03-07-2023 15:34:53	CATHEDRALE
16	Severo Ochoa	True	False	OPEN	Santander	20	20	0	18-12-2023 17:04:43	SEVERO OCHOA
50013	ALLÉE DU BELVÉDÈRE ANGLE ROND POINT DES RAIES - 95000 CERGY	False	False	CLOSED	Cergy-Pontoise	20	20	0	09-06-2025 20:19:22	CER13 - ROND-POINT DES RAIES
50022	BOULEVARD DE L'OISE / RUE DE LA GARE - 95000 CERGY	True	False	OPEN	Cergy-Pontoise	20	20	0	03-06-2025 11:22:35	CER22 - GARE CERGY GRAND CENTI
50005	AVENUE DU HAZAY ANGLE RUE DE L'AISSELETTE - 95000 CERGY	False	False	CLOSED	Cergy-Pontoise	20	20	0	11-06-2025 18:13:19	CER05 - MOULIN A VENT
53002	Place de la Libération côté avenue de Verdun - 95300 PONTOISE	False	False	CLOSED	Cergy-Pontoise	20	20	0	11-06-2025 18:13:53	PON02 - PLACE DE LA LIBÉRATION
50014	AVENUE DU NORD ANGLE ALLÉE DES ACACIAS - 95000 CERGY	False	False	CLOSED	Cergy-Pontoise	20	20	0	11-06-2025 18:16:47	CER14 - AVENUE DU NORD
241	PLAZA NUEVA - Aprox. C/ Mendez Núñez	False	False	OPEN	Seville	20	20	0	23-10-2025 10:04:38	PLAZA NUEVA
4	DEVANT N° 39 ALLEE EUGENE DELACROIX	True	False	CLOSED	Rouen	20	20	0	03-07-2023 14:59:59	MUSEE DES BEAUX ARTS
4	Cormoran (Aprox. C/ de Manuel García Lago)	True	False	CLOSED	Santander	20	20	0	17-01-2024 14:39:26	CORMORÁN
16	AVENUE JACQUES CHASTELLAIN - PARKING DE LA PATINOIRE	True	False	CLOSED	Rouen	20	20	0	03-07-2023 16:16:05	ILE LACROIX
53003	Avenue François Mitterand côté boulevard du Port - 95300 PONTOISE	False	False	CLOSED	Cergy-Pontoise	20	20	0	11-06-2025 18:13:45	PON03 - UNIVERSITÉ ST MARTIN
21	Place Léon Gontier	False	False	OPEN	Amiens	20	20	0	05-06-2025 17:40:27	MAISON DE LA CULTURE
50027	7 Rue des Chauffours, 95000 Cergy	False	False	CLOSED	Cergy-Pontoise	20	20	0	11-06-2025 18:11:07	CER27 - MARJOBERTS
50023	AVENUE DU SUD - 95000 CERGY	False	False	CLOSED	Cergy-Pontoise	20	20	0	13-06-2025 12:42:21	CER23 - LES TOULEUSES
17	Mail Albert 1er	False	False	OPEN	Amiens	20	20	0	28-06-2025 13:50:59	CIRQUE
2012	GRAND RUE - FACE AU 35 GRAND'RUE	True	False	CLOSED	Marseille	20	20	0	19-12-2022 20:02:52	GRAND RUE
11	COURS CLEMENCEAU	True	False	CLOSED	Rouen	20	20	0	03-07-2023 13:29:45	CLEMENCEAU JOFFRE
2	Plaza de las Brisas (Aprox. Jardines de Piquio)	True	False	CLOSED	Santander	20	20	0	17-01-2024 14:39:18	PLAZA DE LAS BRISAS
3	Bd du jardin des plantes	False	False	OPEN	Amiens	20	20	0	10-06-2025 12:58:48	VOGEL
12	PLACE DES COTONNIERS	True	False	OPEN	Rouen	20	20	0	30-06-2022 14:08:00	PLACE DES COTONNIERS
28	Not Found	True	False	CLOSED	Rouen	20	20	0	03-07-2023 15:04:59	BOULINGRIN
53004	BD DE L'OISE ANGLE RUE DE LA PALETTE	False	False	CLOSED	Cergy-Pontoise	20	20	0	12-06-2025 07:07:04	PON04 - LA PALETTE
242	PLAZA NUEVA - Aprox. C/ Mendez Núñez	False	False	OPEN	Seville	20	20	0	23-10-2025 10:05:07	PLAZA NUEVA

Dimension Tables

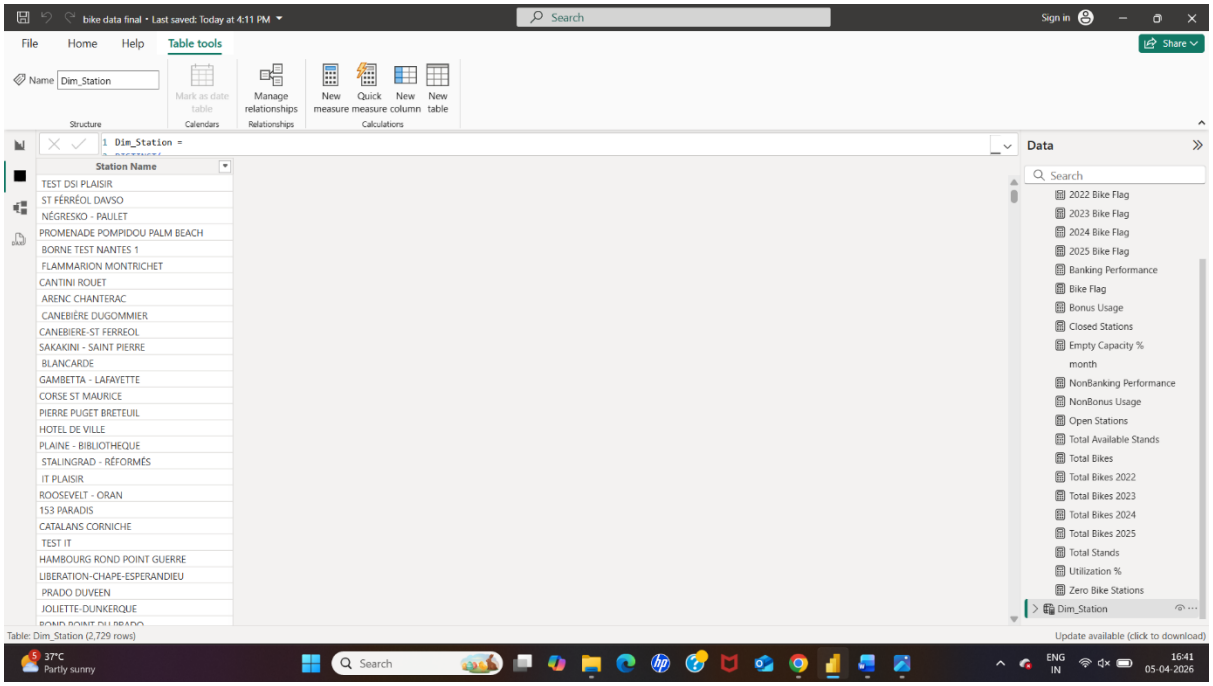
1. Dim_City – Unique list of cities

```
Dim_City =
DISTINCT(
   SELECTCOLUMNS(
      Bike_Data,
      "City Name", Bike_Data[City Name]
   )
)
```



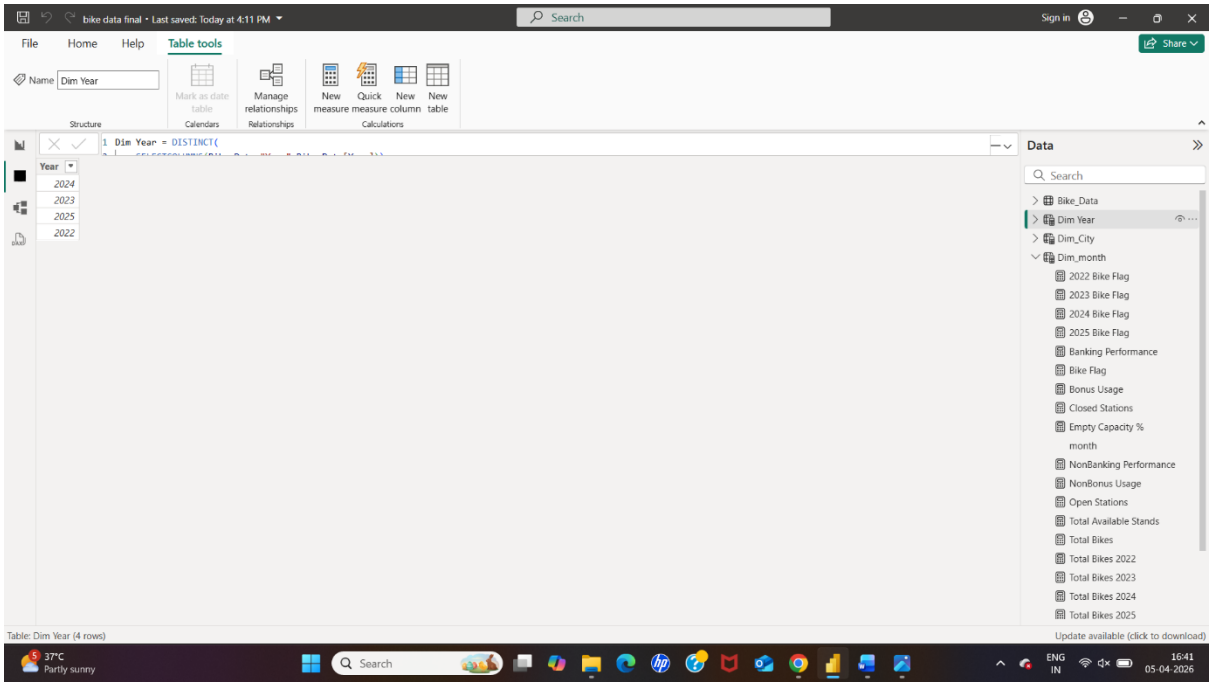
2. Dim_Station – Unique list of stations

```
Dim_Station =
DISTINCT(
    SELECTCOLUMNS(
        Bike_Data,
        "Station Name", Bike_Data[Station Name]


    )
)
```



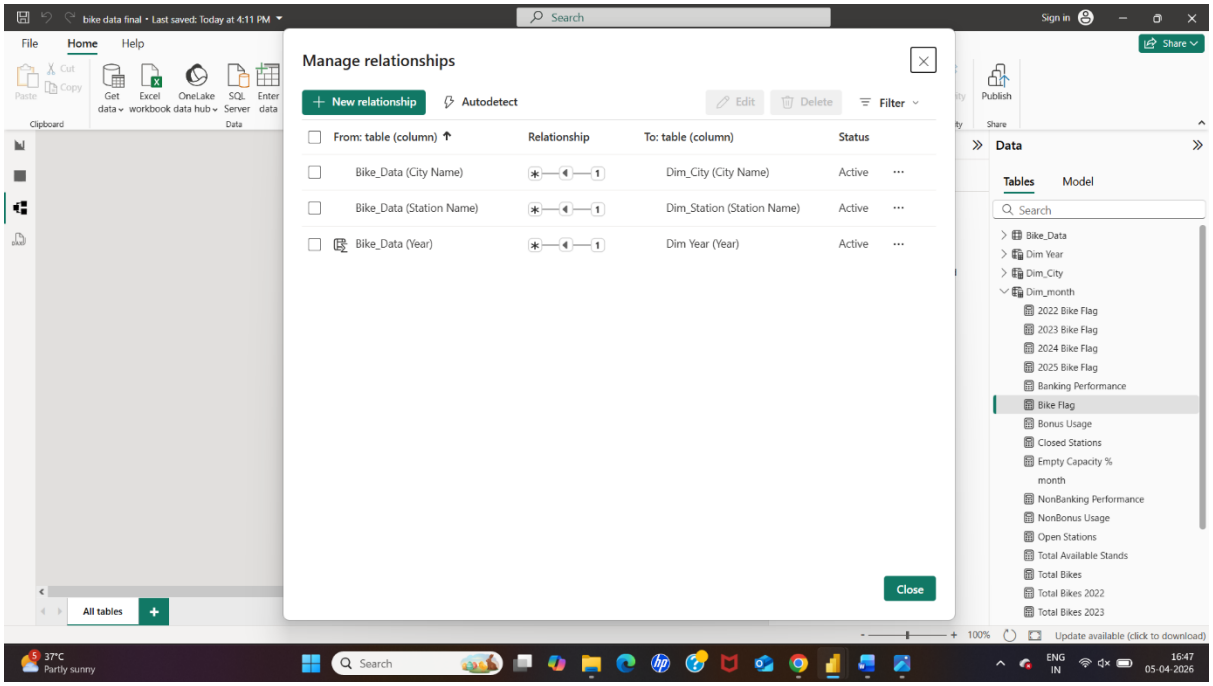
3. Dim_Year – Unique list of years

```
Dim Year = DISTINCT(
    SELECTCOLUMNS(Bike_Data,"Year",Bike_Data[Year]))
```



Relationships

- City Name → One-to-Many
- Station Name → One-to-Many
- Year → One-to-Many



7.Charts and Insights

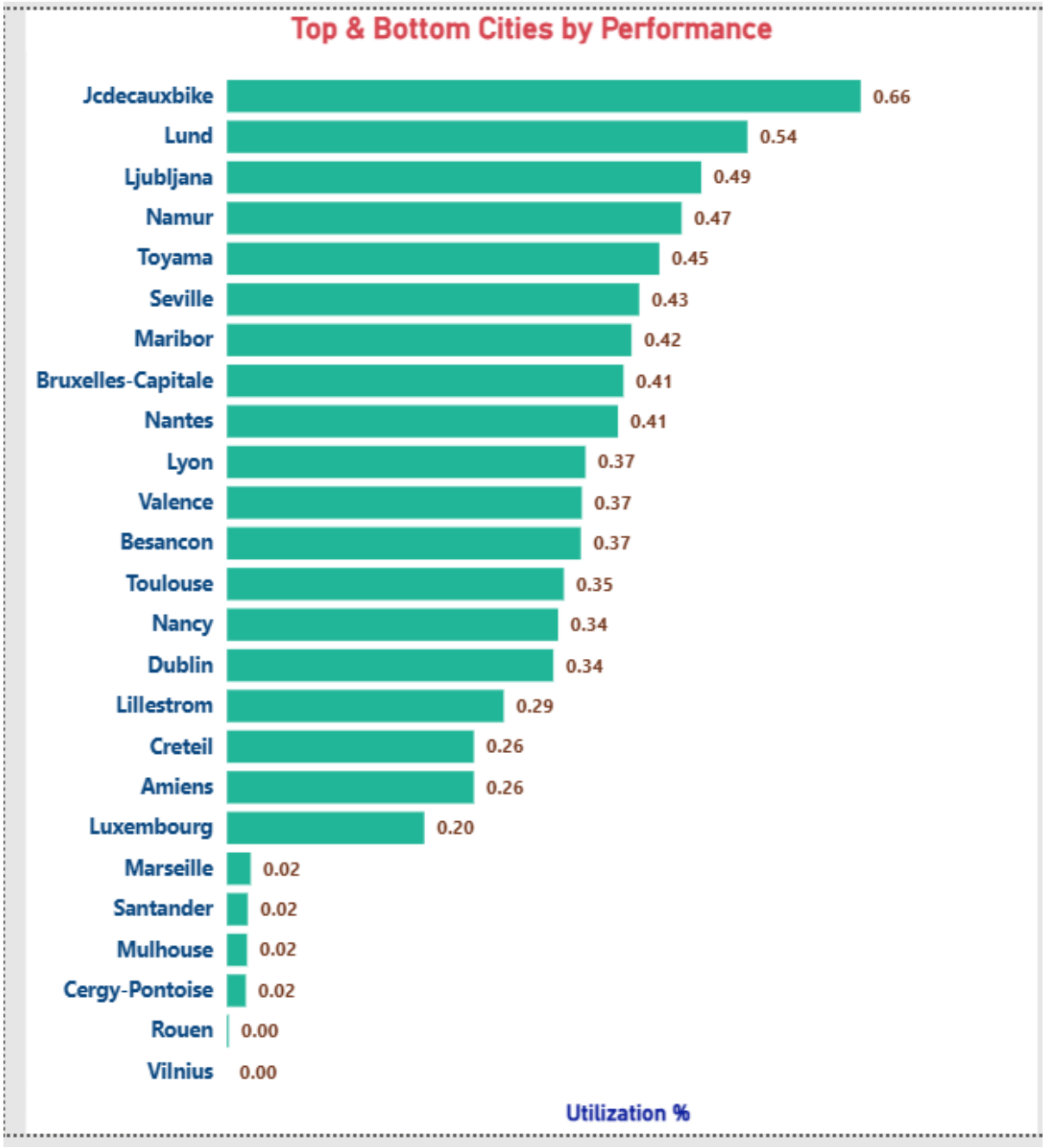
1. Which cities have the highest and lowest bike utilization?

Total Bikes = SUM(Bike_Data[Available Bikes])

Total Stands = SUM(Bike_Data[Bike Stands])

Utilization % =

DIVIDE([Total Bikes], [Total Stands])



Key Insights

- Cities like **Jdeceauxbik**e and **Lund** have high utilization → bikes are used efficiently.
- Cities like **Rouen** and **Vilnius** have very low utilization → poor demand or excess capacity.
- There is a huge gap between top and bottom cities, showing imbalance in usage.
- Some cities may need better bike distribution or demand improvement.

2.How has bike stand availability changed over the years?

Total Available Bike Stands =

SUM(Bike_Data[Available Bike Stands])

Key Insights

- From **2022 to 2024**, available bike stands are low and slightly decreasing, indicating better utilization of stations.
- **I**n **2025**, there is a sharp increase in available bike stands, showing a rise in unused capacity.
- The increase suggests that more stations or docking points were added in **2025**.
- However, the number of bikes did not increase proportionally, leading to more empty stands.
- Higher available stands indicate lower demand or inefficient resource allocation.
- The gap between total capacity and actual usage has widened in **2025**.
- This may result in underutilization of infrastructure if demand does not improve.
- The system may require better planning in terms of station placement and bike distribution.

3.How has bike availability changed across cities over the years, and which cities show the highest growth or decline?

2022 Bike Flag = IF([Total Bikes 2022]=0,"No Bikes","Available")

2023 Bike Flag = IF([Total Bikes 2023]=0,"No Bikes","Available")

2024 Bike Flag = IF([Total Bikes 2024]=0,"No Bikes","Available")

2025 Bike Flag = IF([Total Bikes 2025]=0,"No Bikes","Available")

```
Total Bikes 2025 =
CALCULATE(
    SUM(Bike_Data[Available Bikes]),
    Bike_Data[Year] = 2025
)
Total Bikes 2024 =
CALCULATE(
    SUM(Bike_Data[Available Bikes]),
    Bike_Data[Year] = 2024
)
Total Bikes 2023 =
CALCULATE(
    SUM(Bike_Data[Available Bikes]),
    Bike_Data[Year] = 2023
)
Total Bikes 2022 =
CALCULATE(
    SUM(Bike_Data[Available Bikes]),
    Bike_Data[Year] = 2022
)
```

Name	Total Bikes 2025	2025 Bike Flag
Bruxelles-Capitale	3524	Available
Lyon	3473	Available
Toulouse	2686	Available
Seville	2209	Available
Valence	2012	Available
Dublin	1240	Available
Nantes	1029	Available
Ljubljana	839	Available
Luxembourg	576	Available
Amiens	353	Available
Maribor	328	Available
Lund	276	Available
Nancy	241	Available
Toyama	208	Available
Namur	204	Available
Besancon	175	Available
Lillestrom	34	Available
Jcdecauxbike	17	Available
Cergy-Pontoise	13	Available
Mulhouse	13	Available
Vilnius	0	No Bikes
Creteil		No Bikes
Marseille		No Bikes
Rouen		No Bikes
Santander		No Bikes
Total	**19450**	**Available**

City Name	Total Bikes 2024	2024 Bike Flag
Creteil	60	Available
Seville	48	Available
Bruxelles-Capitale	8	Available
Santander	8	Available
Cergy-Pontoise	4	Available
Jcdecauxbike	3	Available
Lyon	2	Available
Namur	2	Available
Besancon	0	No Bikes
Mulhouse	0	No Bikes
Nantes	0	No Bikes
Toulouse	0	No Bikes
Toyama	0	No Bikes
Amiens		No Bikes
Dublin		No Bikes
Lillestrom		No Bikes
Ljubljana		No Bikes
Lund		No Bikes
Luxembourg		No Bikes
Maribor		No Bikes
Marseille		No Bikes
Nancy		No Bikes
Rouen		No Bikes
Valence		No Bikes
Vilnius		No Bikes
Total	**135**	**Available**

City Name	Total Bikes 2023	2023 Bike Fl
Creteil	4	Available
Jcdecauxbike	2	Available
Dublin	1	Available
Lillestrom	1	Available
Luxembourg	1	Available
Rouen	1	Available
Valence	1	Available
Cergy-Pontoise	0	No Bikes
Lyon	0	No Bikes
Namur	0	No Bikes
Santander	0	No Bikes
Seville	0	No Bikes
Amiens		No Bikes
Besancon		No Bikes
Bruxelles-Capitale		No Bikes
Ljubljana		No Bikes
Lund		No Bikes
Maribor		No Bikes
Marseille		No Bikes
Mulhouse		No Bikes
Nancy		No Bikes
Nantes		No Bikes
Toulouse		No Bikes
Toyama		No Bikes
Vilnius		No Bikes
Total	**11**	**Available**

City Name	Total Bikes 2022	2022 Bike
Marseille	46	Available
Lund	3	Available
Jcdecauxbike	1	Available
Lyon	0	No Bikes
Maribor	0	No Bikes
Nancy	0	No Bikes
Nantes	0	No Bikes
Rouen	0	No Bikes
Seville	0	No Bikes
Amiens		No Bikes
Besancon		No Bikes
Bruxelles-Capitale		No Bikes
Cergy-Pontoise		No Bikes
Creteil		No Bikes
Dublin		No Bikes
Lillestrom		No Bikes
Ljubljana		No Bikes
Luxembourg		No Bikes
Mulhouse		No Bikes
Namur		No Bikes
Santander		No Bikes
Toulouse		No Bikes
Toyama		No Bikes
Valence		No Bikes
Vilnius		No Bikes
Total	**50**	**Available**

Key Insights

1. Strong Growth Cities

Bruxelles-Capitale

- 2022: No bikes
- 2023: No bikes
- 2024: Small increase
- 2025: Huge jump (3524)
- Shows major expansion and strong growth

Lyon

- 2022: 0 bikes
- 2023: 0 bikes
- 2024: Low (2)
- 2025: Very high (3473)
- Indicates sudden development in 2025

2. Consistently Improving Cities

Toulouse, Seville, Valencia

- Very low or no bikes in early years
- Gradual increase
- Strong performance in 2025
- Shows steady improvement

3. Moderate Growth Cities

Dublin, Nantes, Ljubljana

- Small values in 2023–2024
- Good increase in 2025
- Growing but not top performers

4. Low Performing Cities

Cergy-Pontoise, Mulhouse

- Very low numbers even in 2025
- Indicates weak demand or small system

5. No Growth / Poor Cities

Rouen, Vilnius

- Almost zero across all years
- No improvement

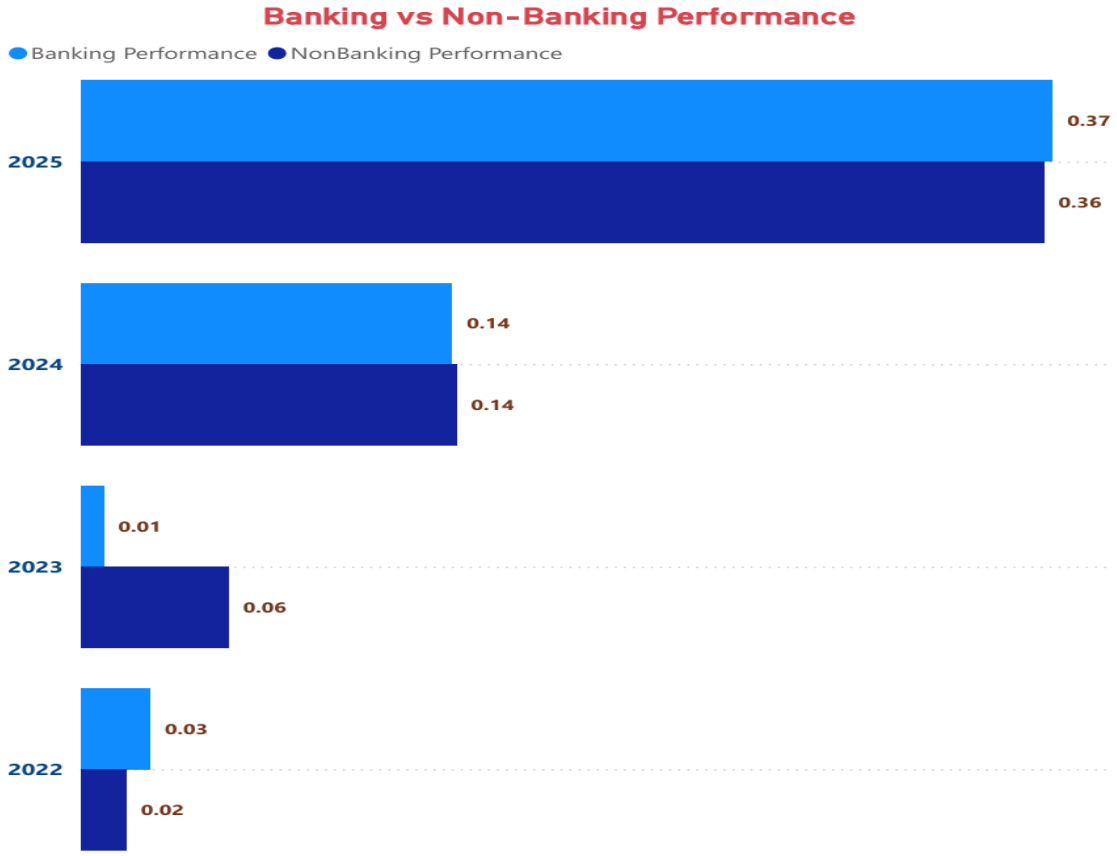
- Indicates inactive or failed locations

6. Overall Trend

- 2022 → Very low availability (initial stage)
- 2023 → Slight improvement
- 2024 → Small growth
- 2025 → Massive increase across most cities
- System moved from early stage to rapid expansion phase

4.How does performance differ between banking and non-banking stations over the years?

```
Banking Performance =

CALCULATE(

    DIVIDE(SUM(Bike_Data[Available Bikes]), SUM(Bike_Data[Bike Stands])),

    Bike_Data[Banking] = TRUE()

)

Non-Banking Performance =

CALCULATE(

    DIVIDE(SUM(Bike_Data[Available Bikes]), SUM(Bike_Data[Bike Stands])),

    Bike_Data[Banking] = FALSE()
```



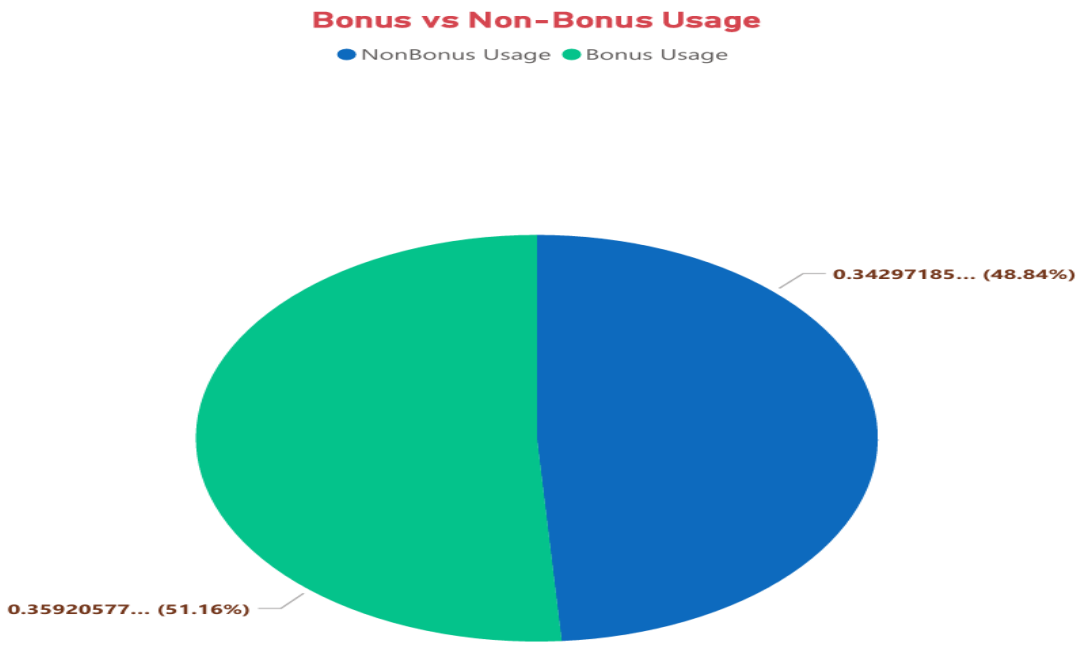
Key Insights

- In 2022, banking stations perform **slightly better** than non-banking stations, though overall utilization is very low.
- In 2023, non-banking stations show higher performance compared to banking stations, indicating a temporary shift in usage.
- In 2024, both banking and non-banking stations perform equally, showing balanced utilization.
- In 2025, both categories show significant improvement, with banking stations slightly outperforming non-banking stations.
- Overall, the performance gap between banking and non-banking stations is minimal, indicating that banking facility does not create a major difference in usage.
- The major improvement in 2025 suggests overall system growth rather than influence from banking feature.

5.Does the bonus feature significantly impact bike usage?

```
Bonus Usage =

CALCULATE(

    SUM(Bike_Data[Available Bikes]),

    Bike_Data[Bonus] = TRUE()
```

```
Non-Bonus Usage =

CALCULATE(

    SUM(Bike_Data[Available Bikes]),

    Bike_Data[Bonus] = FALSE()

)
```



Key Insights

- Bonus usage accounts for **approximately 51%** of total usage, while non-bonus usage accounts for around **49%.**
- The distribution between bonus and non-bonus usage is almost equal, indicating no strong dominance.
- The slight advantage of bonus usage suggests that incentives may have a small positive impact on usage.
- However, the difference is not significant, meaning bonus features alone do not strongly influence user behaviour.
- Usage patterns appear to be driven more by availability and location rather than bonus incentives.

6.How are bike resources distributed geographically across different cities, and which locations show higher availability?

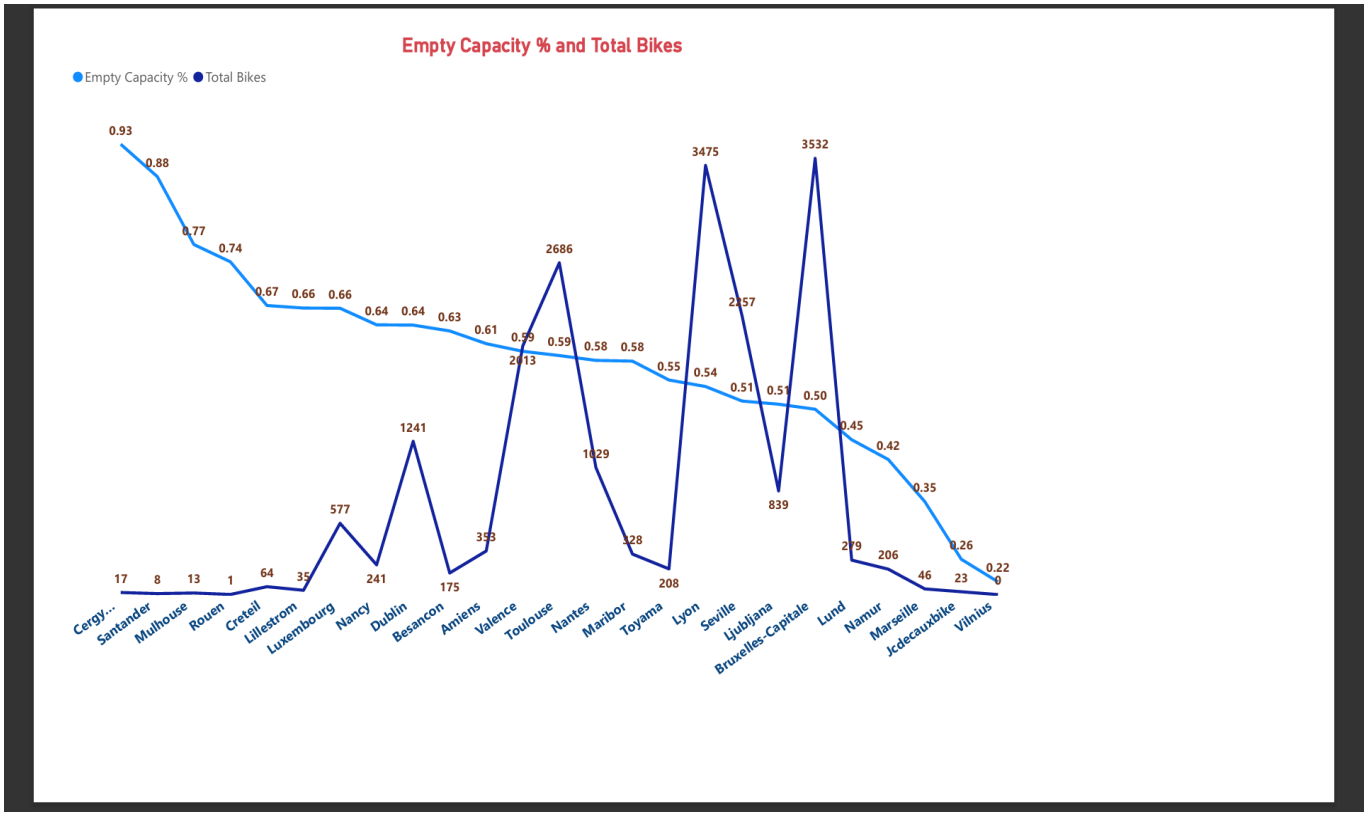
```
Total Bikes 2025 =

CALCULATE(

    SUM(Bike_Data[Available Bikes]),

    YEAR(Bike_Data[Last Update]) = 2025

)
```

Key Insights

- Cities in **France** region (like Lyon, Bruxelles-Capitale, Toulouse) show high bike availability, indicating strong operational presence.
- **Bruxelles-Capitale** and **Lyon** have the largest bubbles, meaning they are top contributors in bike availability.
- Central European cities such as **Ljubljana** and **Lun**d show moderate availability, indicating balanced but smaller systems.
- Some cities like **Vilnius** and smaller regions have very low or almost no availability, showing weak or inactive networks.
- The map clearly shows that bike distribution is clustered in specific regions rather than evenly spread.
- **Western Europe** has higher density compared to Northern and Eastern regions.
- This uneven distribution suggests that expansion opportunities exist in underperforming or low-density areas.

7.How does empty capacity vary across cities, and how is it related to the total number of available bikes?

```
Total Bikes =

SUM(Bike_Data[Available Bikes])

Empty Capacity % =

DIVIDE(

    SUM(Bike_Data[Available Bike Stands]),

    SUM(Bike_Data[Bike Stands])

)
```

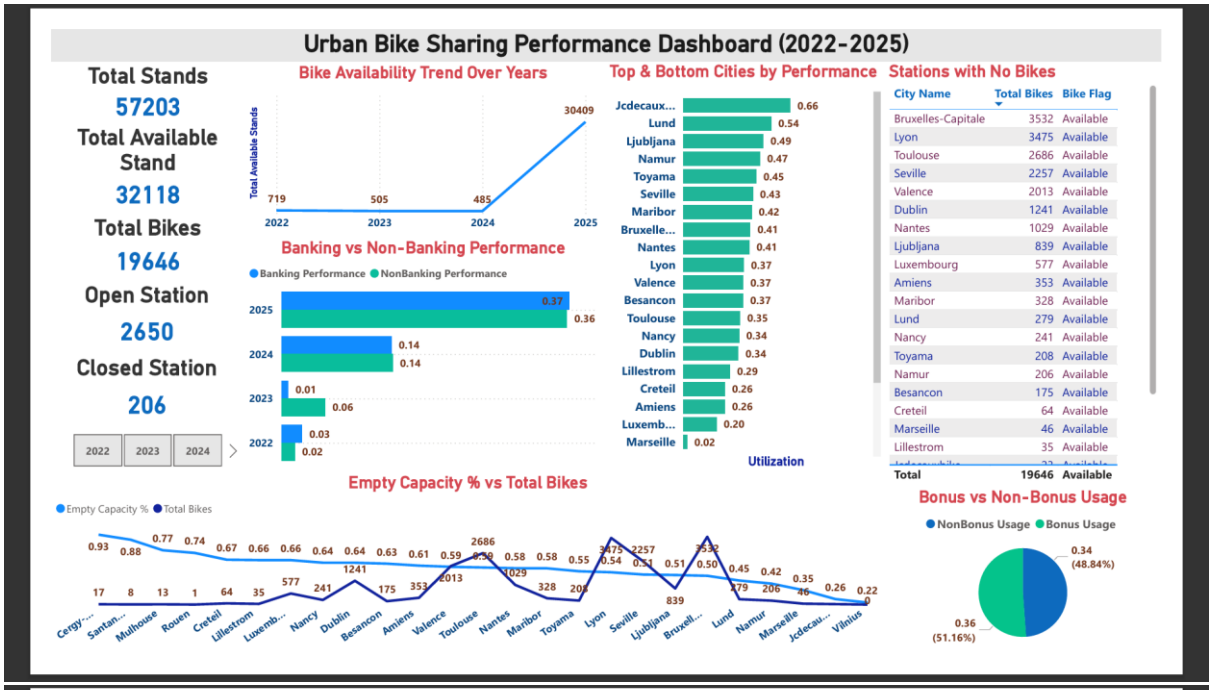


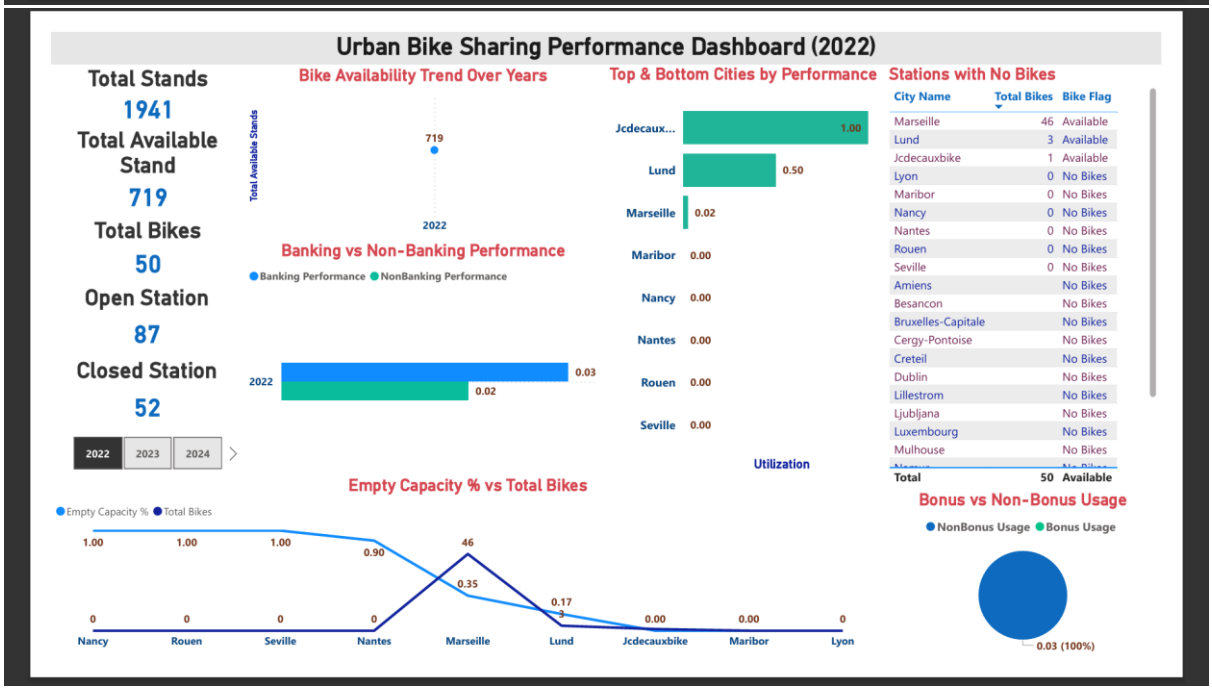
Key Insights

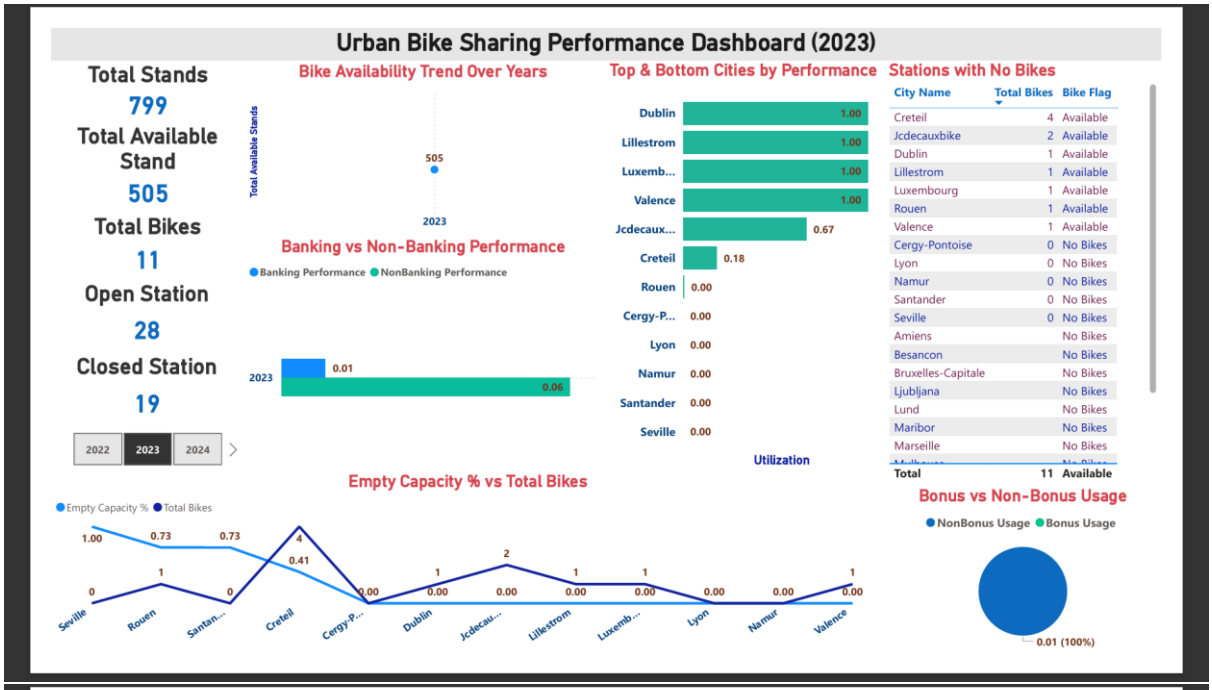
- Cities like **Cergy-Pontoise, Santander, and Mulhouse** show very high empty capacity (above 80%), indicating that most bike stands are empty and demand is extremely low.
- **Rouen, Creteil, and Lillestrom** also have high empty capacity with very low bike availability, suggesting underutilized stations.
- Mid-level cities such as **Nancy, Dublin, Besancon, and Amiens** show moderate empty capacity with gradual improvement in bike availability.
- High-performing cities like **Lyon, Bruxelles-Capitale, and Toulouse** have a strong number of bikes but still maintain moderate empty capacity, indicating balanced usage.

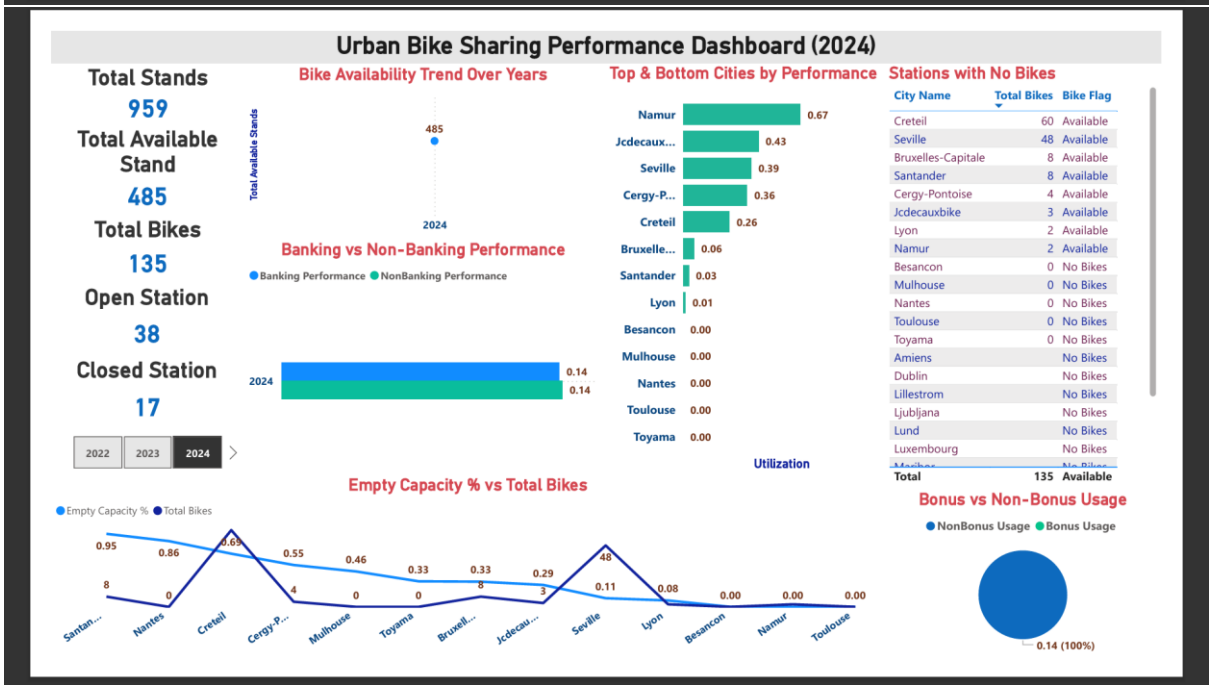
- **Seville and Ljubljana** show good bike availability with relatively lower empty capacity, meaning better utilization efficiency.
- Cities like **Marseille, Jcdecauxbike, and Vilnius** have low empty capacity but also very low total bikes, which may indicate smaller systems rather than high efficiency.
- There is an overall inverse pattern where cities with higher bikes tend to reduce empty capacity, but not always, showing operational differences.

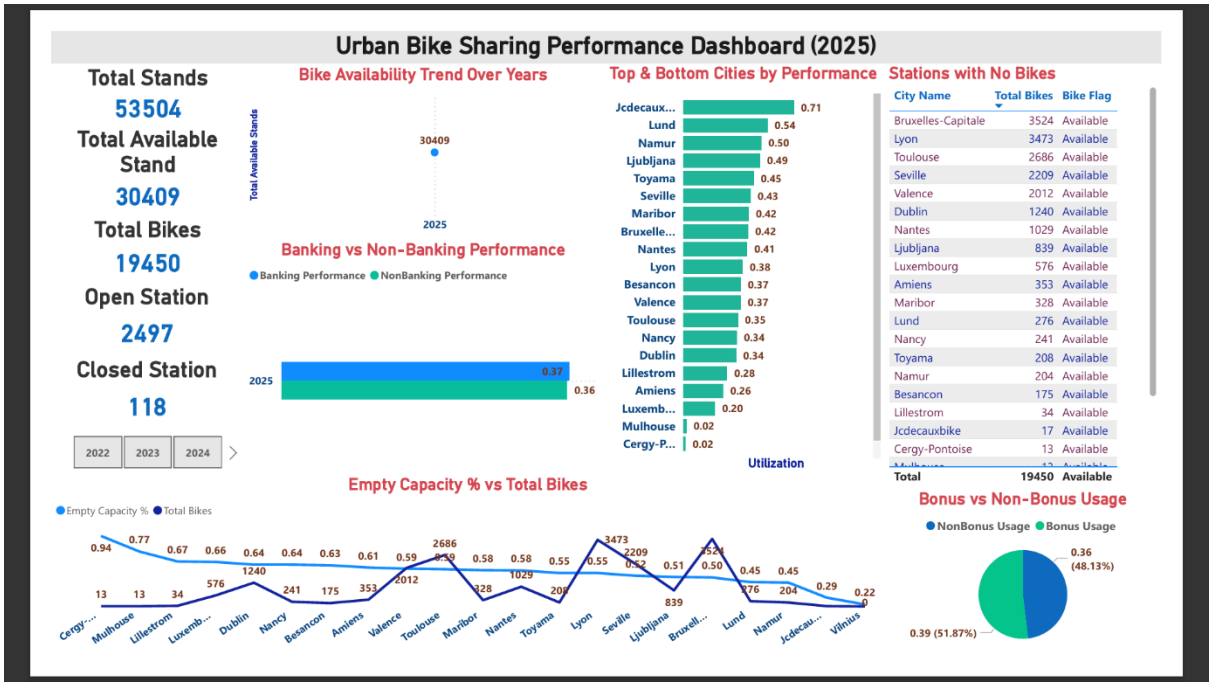
8.Final Dashboard






Urban Bike Sharing Performance Dashboard (2023)
Total Stands
799
Total Available Stand
505
Total Bikes
11
Open Station
28
Closed Station
19
2022
2023
2024
Bike Availability Trend Over Years
Total Available Stands
505
2023
Banking vs Non-Banking Performance
Banking Performance
NonBanking Performance
2023
0.01
0.06
Top & Bottom Cities by Performance
Dublin 1.00
Lillestrom 1.00
Luxemb... 1.00
Valence 1.00
Jcdecaux... 0.67
Creteil 0.18
Rouen 0.00
Cergy-P... 0.00
Lyon 0.00
Namur 0.00
Santander 0.00
Seville 0.00
Utilization
Stations with No Bikes
City Name Total Bikes Bike Flag
Creteil 4 Available
Jcdecauxbike 2 Available
Dublin 1 Available
Lillestrom 1 Available
Luxembourg 1 Available
Rouen 1 Available
Valence 1 Available
Cergy-Pontoise 0 No Bikes
Lyon 0 No Bikes
Namur 0 No Bikes
Santander 0 No Bikes
Seville 0 No Bikes
Amiens No Bikes
Besancon No Bikes
Bruxelles-Capitale No Bikes
Ljubljana No Bikes
Lund No Bikes
Maribor No Bikes
Marseille No Bikes
Total 11 Available
Empty Capacity % vs Total Bikes
Empty Capacity %
Total Bikes
Seville Rouen Santan... Creteil Cergy-P... Dublin Jcdecau... Lillestrom Luxemb... Lyon Namur Valence
Bonus vs Non-Bonus Usage
NonBonus Usage
Bonus Usage
0.01 (100%)


Urban Bike Sharing Performance Dashboard (2024)
Total Stands
959
Total Available Stand
485
Total Bikes
135
Open Station
38
Closed Station
17
2022
2023
2024
Bike Availability Trend Over Years
Total Available Stands
485
2024
Banking vs Non-Banking Performance
Banking Performance
NonBanking Performance
2024
0.14
0.14
Top & Bottom Cities by Performance
Namur 0.67
Jcdecaux... 0.43
Seville 0.39
Cergy-P... 0.36
Creteil 0.26
Bruxelle... 0.06
Santander 0.03
Lyon 0.01
Besancon 0.00
Mulhouse 0.00
Nantes 0.00
Toulouse 0.00
Toyama 0.00
Utilization
Stations with No Bikes
City Name Total Bikes Bike Flag
Creteil 60 Available
Seville 48 Available
Bruxelles-Capitale 8 Available
Santander 8 Available
Cergy-Pontoise 4 Available
Jcdecauxbike 3 Available
Lyon 2 Available
Namur 2 Available
Besancon 0 No Bikes
Mulhouse 0 No Bikes
Nantes 0 No Bikes
Toulouse 0 No Bikes
Toyama 0 No Bikes
Amiens No Bikes
Dublin No Bikes
Lillestrom No Bikes
Ljubljana No Bikes
Lund No Bikes
Luxembourg No Bikes
Total 135 Available
Empty Capacity % vs Total Bikes
Empty Capacity %
Total Bikes
Santan... Nantes Creteil Cergy-P... Mulhouse Toyama Bruxell... Jcdecau... Seville Lyon Besancon Namur Toulouse
Bonus vs Non-Bonus Usage
NonBonus Usage
Bonus Usage
0.14 (100%)



Key Insights

Descriptive Analysis

From **2022 to 2025**, the bike-sharing system shows a clear transformation. In **2022**, the system was very small with only **50 bikes and limited active cities**, and most stations had no bikes. In **2023**, the situation remained weak, dropping further to **11 bikes**, indicating almost no operational activity.

In **2024**, the system started recovering with **135 bikes and slight city-level improvements**, but growth was still limited. The major change happens in **2025**, where total bikes increased drastically to **19,450**, along with **30,409 available stands**, showing a large-scale expansion.

City-wise, only a few cities were active in early years, but by 2025, major cities like **Bruxelles-Capitale, Lyon, Toulouse, and Seville** dominate the system. Bonus usage also becomes visible only in 2025, contributing more than non-bonus usage.

Diagnostic Analysis

The low performance in **2022 and 2023** is mainly due to **lack of infrastructure, fewer stations, and low adoption across cities**, as seen from high empty capacity and many "No Bikes" stations.

The slight improvement in **2024** indicates **initial system expansion**, where some cities like **Namur and Seville** begin to show activity. However, the system was still not fully developed.

The massive growth in **2025** is driven by **significant expansion of stations and bikes**, along with **better distribution across cities**. High-performing cities like **Bruxelles-Capitale and Lyon** indicate strong urban demand and better operational planning.

Additionally, the introduction of **bonus usage** helped increase user engagement, and the reduction in empty capacity shows **improved utilization efficiency**.

Predictive Analysis

If the current trend continues, the system is expected to **grow further beyond 2025**, with more cities becoming active and existing cities expanding their capacity.

Top-performing cities like **Bruxelles-Capitale and Lyon** are likely to remain dominant and may require further infrastructure scaling to meet demand. Moderate cities such as **Dublin, Nantes, and Ljubljana** are expected to move into higher performance categories.

Low-performing cities like **Vilnius and Cergy-Pontoise** may either improve with investment or remain inactive if demand does not increase.

Bonus-based usage is expected to increase further, as it has already shown a positive impact in 2025.

Prescriptive Analysis

To improve the system further, focus should be on **expanding infrastructure in high-demand cities** like **Lyon and Bruxelles-Capitale** to avoid capacity shortages.

Moderate cities such as **Dublin and Nantes** should be supported with **more stations and bikes** to accelerate their growth.

For low-performing cities like **Vilnius and Mulhouse**, authorities should **analyze demand patterns** and either improve awareness and accessibility or reconsider resource allocation.

Increasing **bonus incentives** can further boost user engagement, as it has already proven effective in 2025. Additionally, maintaining an optimal balance between **available stands and bikes** will help reduce empty capacity and improve overall efficiency.

9.Conclusion

From **2022 to 2025**, the bike-sharing system evolved from a **low-usage and underdeveloped network** into a **high-growth and well-established system**. The early years (**2022–2023**) showed very low bike availability and poor utilization, while **2024** marked the beginning of recovery.

A major transformation occurred in **2025**, with a significant increase in bikes, stations, and active cities. Key cities like **Bruxelles-Capitale and Lyon** drove most of the growth, and the introduction of **bonus usage** improved user engagement.

Overall, the project highlights that **infrastructure expansion, strong city-level performance, and incentive strategies** are essential for building an efficient and scalable bike-sharing system.